AI BASICS 102

Agent Harnesses

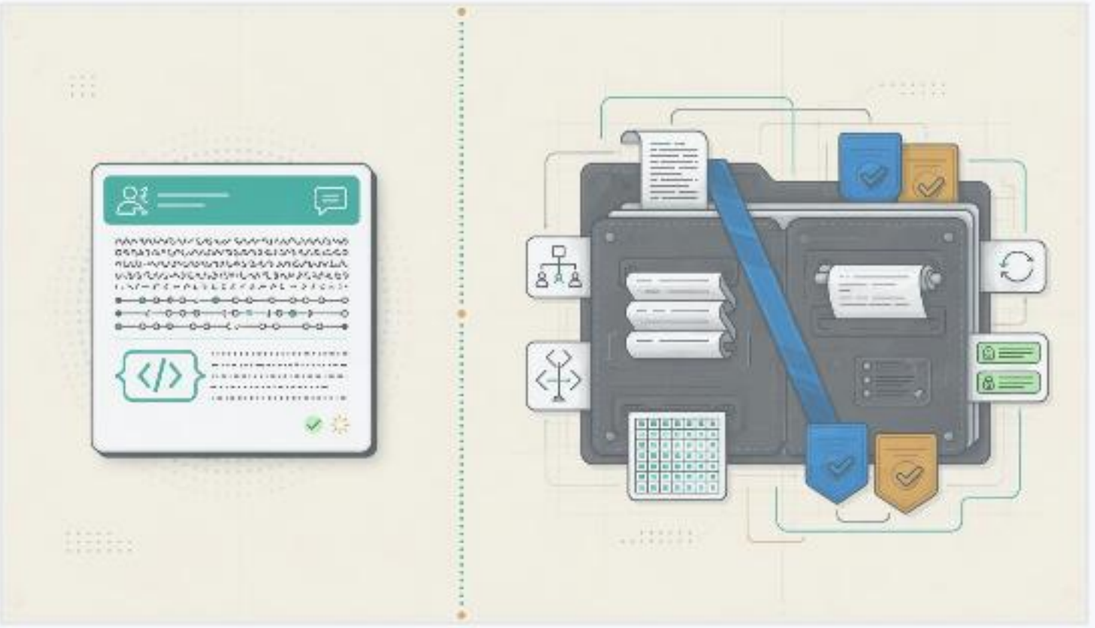
How teams keep Copilot useful, bounded, and reviewable.

Rules | Moves | Tools | Checks

20 minutes | VS Code + GitHub Copilot framing

A good prompt helps once. A harness helps every session.

The repo should carry the rules, checks, and habits the agent should not rediscover.



One session

Prompt, context, and steering.

Every session

Instructions, skills, scripts, checks, and review habits.

The goal is not more trust. The goal is more inspectable work.

Use four pieces: rules, moves, tools, checks.

Rules

Standing instructions and boundaries.

Moves

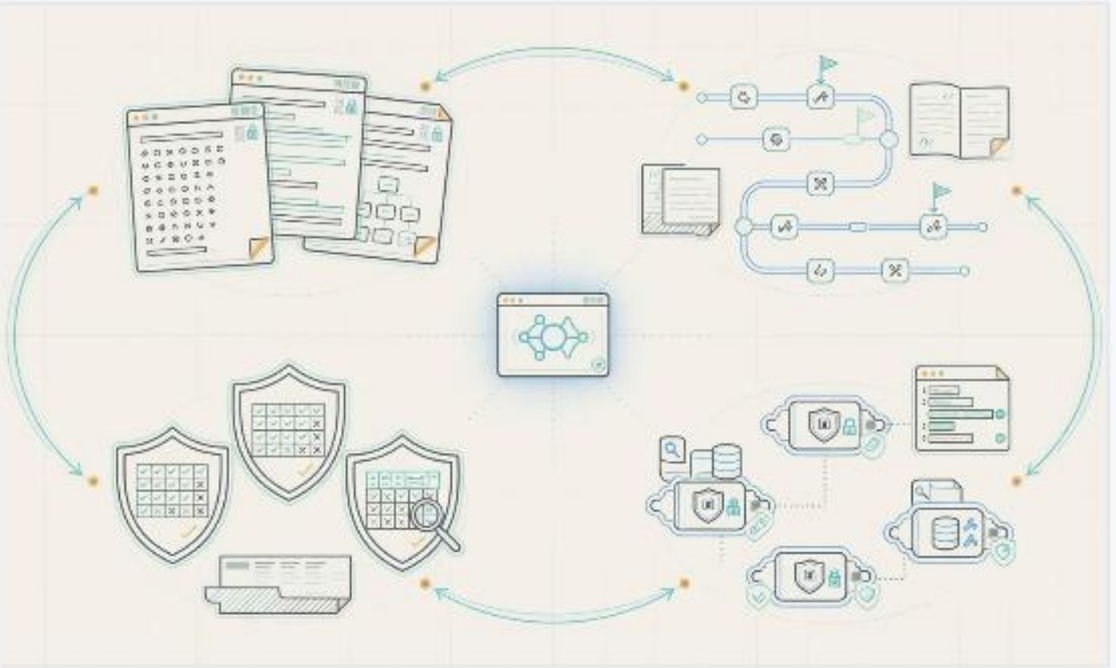
Skills, playbooks, and team procedures.

Tools

Scripts, commands, and controlled capabilities.

Checks

Tests, lint reports, diagnostics, and review.

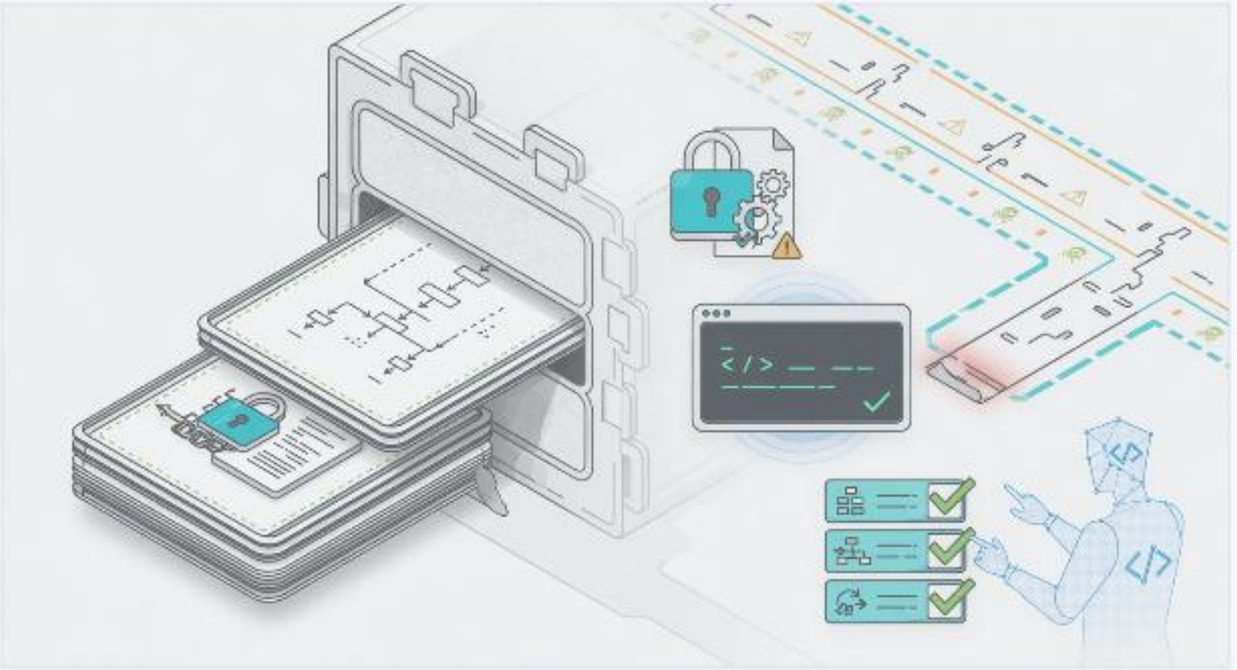


A harness is the system around Copilot, not one magic prompt.

Repo instructions are standing orders.

.github/copilot-instructions.md

Write the rules the agent should not guess every time.



Commands

Package manager, install, build, lint, test, and safe local checks.

Conventions

Architecture patterns, preferred helpers, naming, errors, and logging.

Boundaries

Generated files, risky paths, public APIs, secrets, migrations, and deploys.

Done

What must be verified before Copilot says the work is complete.

VS Code Docs: Copilot custom instructions | GitHub Docs: customizing Copilot responses

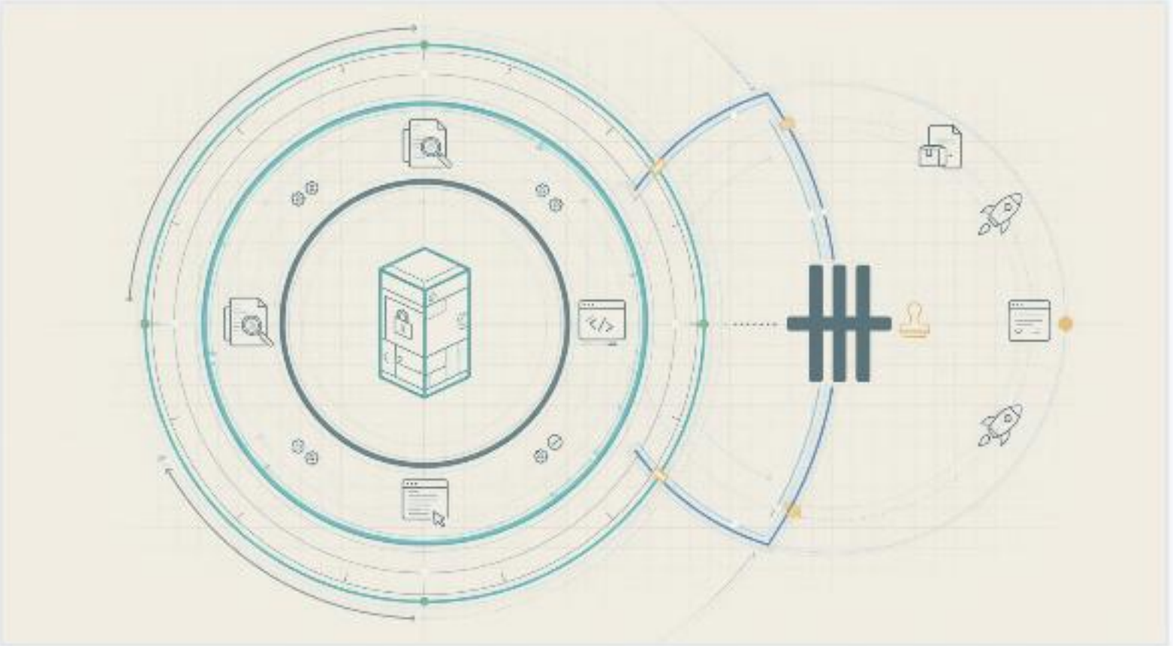
Different actions deserve different permission.

Read
Files, docs, logs, issues.

Edit
Local code, tests, docs.

Run
Allowlisted commands.

Ship
Deploys, data, merges.



Allowed

Read code, inspect tests, update focused local files.

Ask first

Install packages, delete files, run migrations, change public APIs.

Never alone

Touch secrets, production data, releases, or protected branches.

Boundaries make Copilot safer and make the search space smaller.

Skills teach repeatable moves.

Instructions define the rules. Skills define the moves.

Before coding

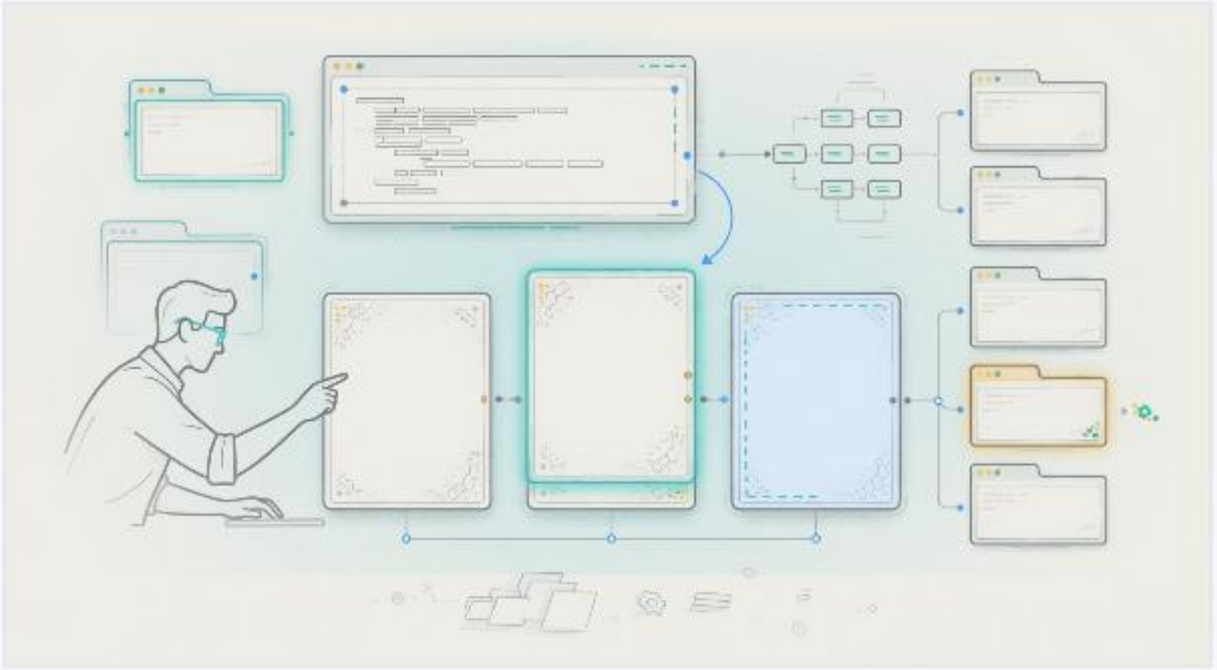
Stress-test the plan, scope, and missing evidence.

After coding

Diagnose the generated feature and hunt for bugs.

During review

Check auth, UI, migrations, or release notes with the team recipe.



grill-me

diagnose

review-auth

ui-smoke

GitHub Docs: Copilot customization cheat sheet | Matt Pocock skills repo: diagnose skill

A harness captures how your team works.

Bug triage

Start with evidence, owner, repro, and one likely surface.

Migration safety

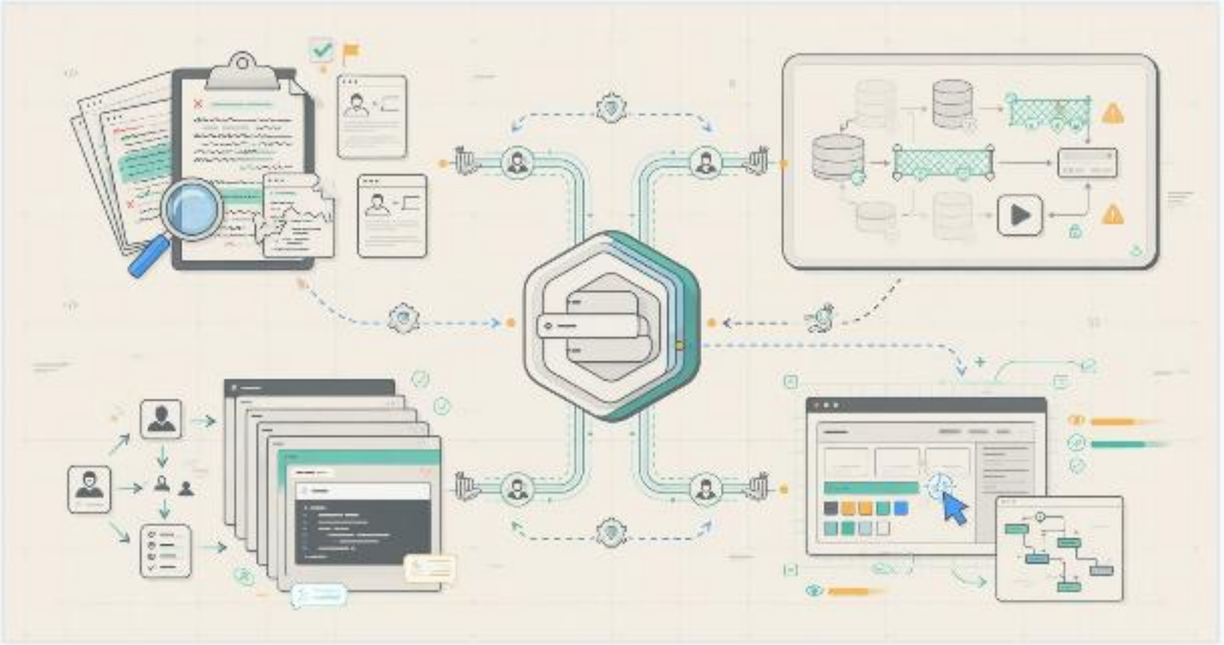
Dry-run first. Human approval before data-changing commands.

UI testing

Run the app, inspect the screen, capture the visual result.

PR review

Summarize evidence, changed behavior, checks, and remaining risk.

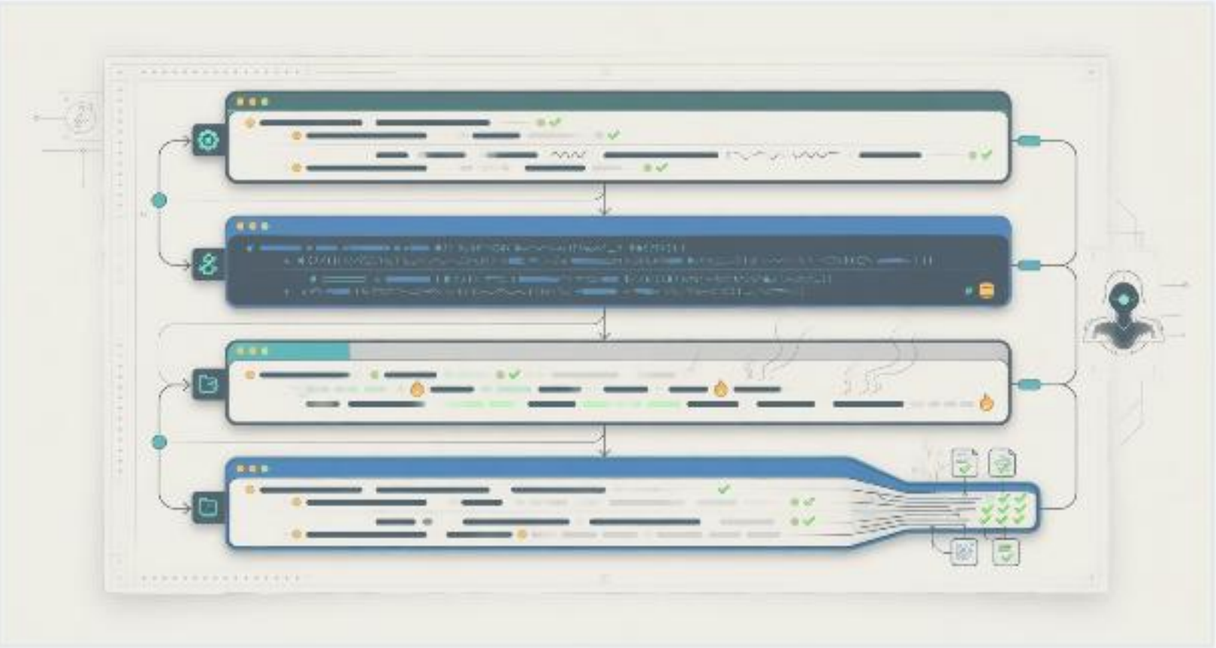


If the team repeats a move, make it part of the harness.

Give Copilot safe paths through verification.

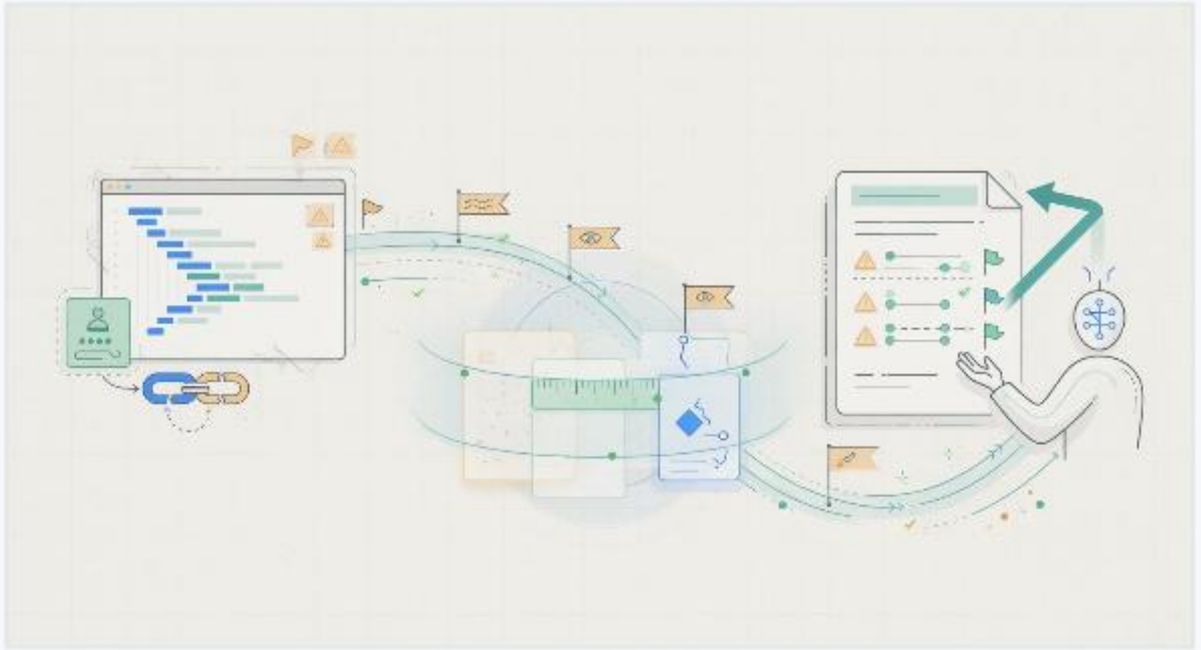
A named script is better than asking the agent to invent the right command.

`npm run test:changed`	Run only tests related to changed files.
`npm run agent:verify`	Bundle typecheck, lint report, focused tests, and smoke checks.
`npm run smoke:local`	Open the app path and confirm the main behavior still works.
`npm run check:pr`	Collect the review evidence a human needs before merging.



Warnings can be a useful middle ground.

Not every repo has hard lint gates.
Agents still need feedback.



Hard gate

Blocks merge when the repo already enforces the rule.

Soft report

Returns warnings so Copilot can fix obvious issues without stopping the whole session.

unused code

unsafe any

accessibility hints

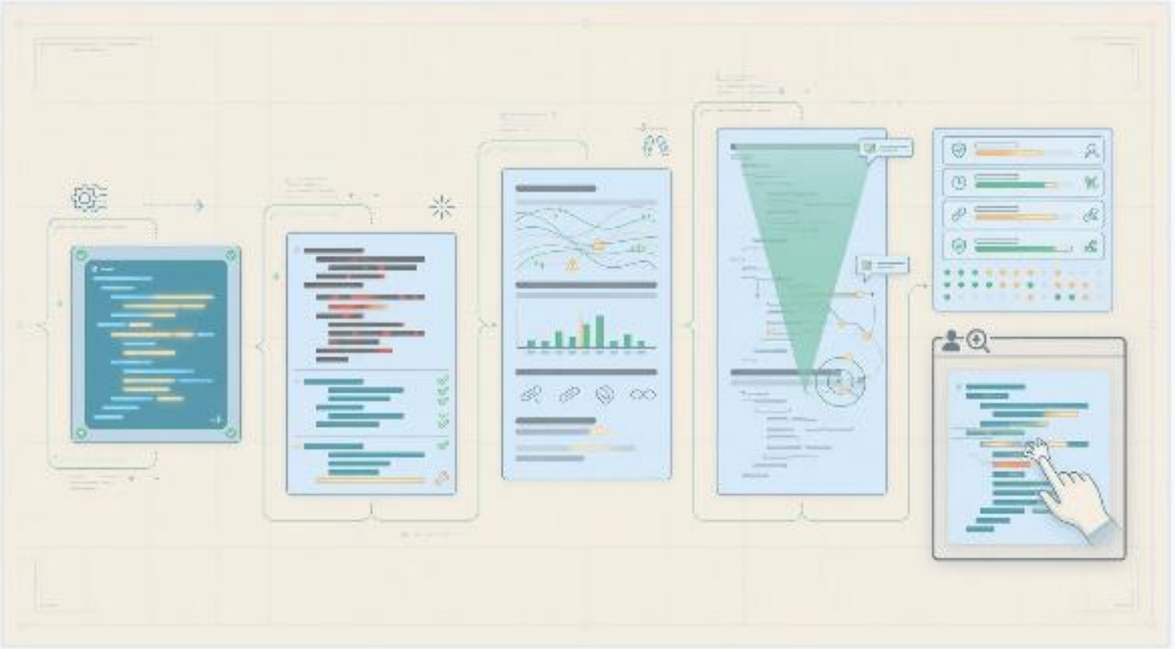
slow tests

wide diff

A soft check is not permission to ignore quality. It is a way to surface quality earlier.

Inspect before review.

Human review starts from evidence, not from a confident summary.



Generate
Copilot edits code inside the task boundary.

>

Run checks
Changed tests, soft lint, typecheck, and smoke commands.

>

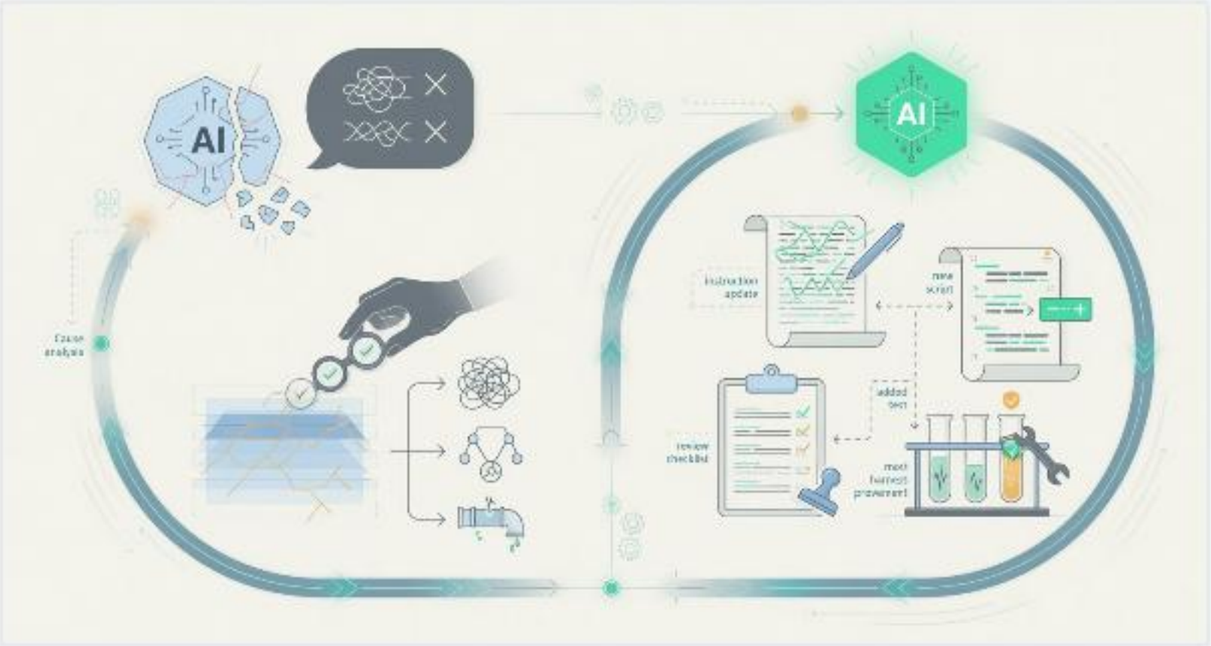
Diagnose
Use a review or debug skill to look for likely bugs.

>

Report
Evidence, changes, checks, gaps, and risk.

When Copilot fails, improve the harness.

Failure > Cause > Harness fix > Next session



Missing context
Add an instruction or path rule.

Wrong command
Add or document a script.

Weak proof
Add a focused test or smoke check.

Bad review
Add a checklist or diagnostic skill.

Do not only fix the prompt. Fix the system that shaped the prompt.

CONCLUSION

Make AI work inspectable.

Bounded

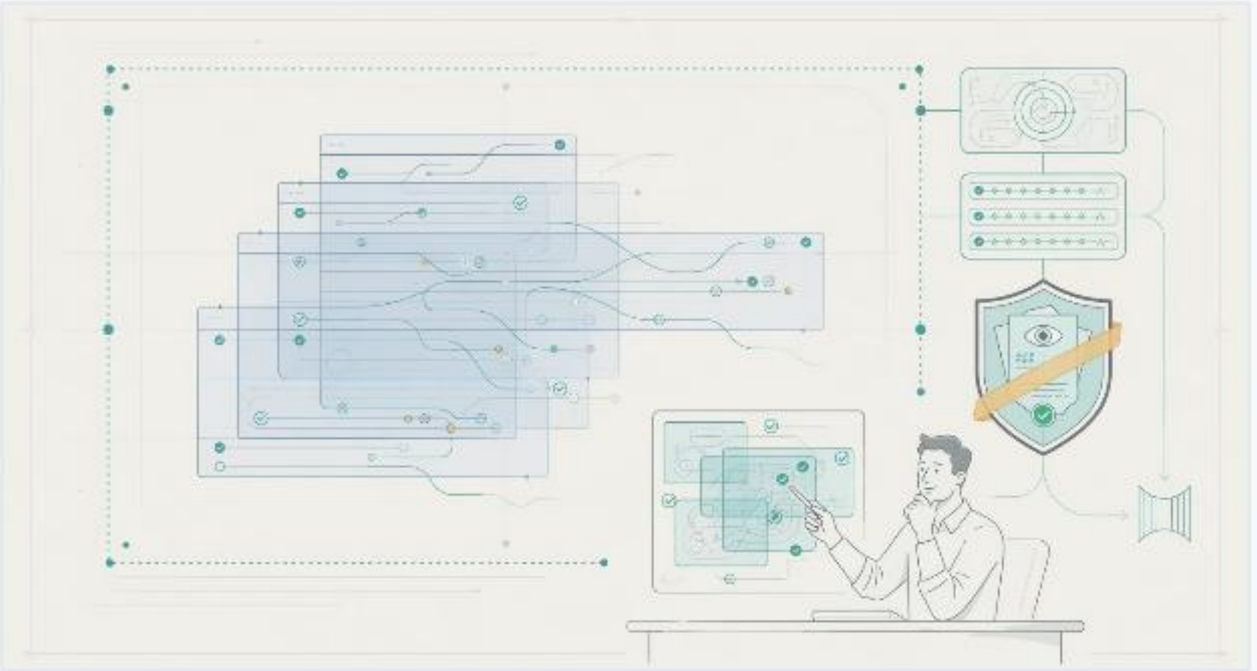
Instructions and permissions define the work area.

Visible

Skills and scripts expose the path the agent followed.

Checkable

Tests, soft reports, diagnostics, and review prove the work.



The goal is not to trust Copilot more. The goal is to make the work easier to inspect.

PRESENTER APPENDIX

Fidelity Tasks VS Code demo cue

Open the Fidelity Tasks repo and show the harness parts in this order.

Rules

Copilot instructions and path-specific boundaries.

Moves

Skills/playbooks such as planning, grilling, diagnosis, and review.

Tools

Changed-test script, heavy soft-lint/report script, and smoke command.

Checks

Post-generation diagnostic pass and human review checklist.

Presenter-only slide. Not included in Ara narration.